Exp: 2 **Study of Network Cables**

Aim:

To understand the different types of network cables and to make a straight-through and a cross-wired cable using a crimping tool.

Requirements:

* UTP cable (Cat5e / Cat6)
* RJ45 connectors
* Crimping tool
* Wire stripper / cutter
* LAN cable tester

Types of Network Cables:

* UTP (Unshielded Twisted Pair): most common cable for LAN and Ethernet.
* STP (Shielded Twisted Pair): has a shield to reduce electromagnetic interference.
* Coaxial cable: used for cable TV and older networks.
* Fibre optic cable: carries data as light; very high speed over long distances.

T568B Colour Code (pin 1 to 8):

White/Orange, Orange, White/Green, Blue, White/Blue, Green, White/Brown, Brown

Cable Types:

* Straight-through: T568B at both ends (PC to switch/hub).
* Cross-wired: T568B at one end, T568A at the other (PC to PC). T568A order: White/Green, Green, White/Orange, Blue, White/Blue, Orange, White/Brown,

Brown.

Procedure:

1. Cut the cable to the required length.
2. Strip about 2-3 cm of the outer insulation.
3. Untwist the pairs and arrange the wires in the required order (T568B / T568A).
4. Trim the wires evenly.
5. Insert the wires fully into the RJ45 connector.
6. Crimp the connector using the crimping tool.
7. Repeat for the other end and test the cable with the LAN tester.

Result:

The straight-through and cross-wired cables were made using the crimping tool and tested successfully.